



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

BT2001-1 – ANALYTICAL TECHNIQUES

(BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **The closeness of the obtained result to the expected result is**
 - A) Precision
 - B) Detection limit
 - C) Accuracy
 - D) Sensitivity
2. **Blood glucose sensor contains which of the following enzyme?**
 - A) Glucose oxidase
 - B) Glucose peroxidase
 - C) Invertase
 - D) Glycogen hydrogenase
3. **Units obtained from basic units are referred as**
 - A) Fundamental units
 - B) Derived units
 - C) Rare units
 - D) Physical units
4. **Buffers are solutions that can resist the change in**
 - A) Temperature
 - B) Volume
 - C) Pressure
 - D) pH
5. **The wavelength corresponding to UV region is**
 - A) 200-400 nm
 - B) 400-800 nm
 - C) 550-750 nm
 - D) 800-900 nm
6. **AAS involves**
 - A) Light emission
 - B) Sample volatilization
 - C) Sample pressurization
 - D) Light scattering
7. **UV-Visible spectroscopy cannot be used for which of the below applications?**
 - A) DNA identification
 - B) Protein identification
 - C) Studying protein denaturation
 - D) Determining protein mass
8. **Which of the following components in a spectrophotometer is used for filtering unwanted radiations?**
 - A) Source
 - B) Monochromator
 - C) Sample cell
 - D) Detector
9. **A non-linear molecule containing n atoms exhibits the following number of vibrations**
 - A) $3n-6$
 - B) $3n-5$
 - C) $3n-4$
 - D) $3n-3$
10. **Which of the following is not a form of molecular vibration?**
 - A) Scissoring
 - B) Twisting
 - C) Wagging
 - D) Hopping
11. **MRI is an application of**
 - A) NMR spectroscopy
 - B) UV-Vis spectroscopy
 - C) IR spectroscopy
 - D) Mass spectroscopy
12. **Which of the following atoms cannot be studied using NMR?**
 - A) ^{13}C
 - B) ^{30}P
 - C) ^1H
 - D) ^{31}P

P.T.O.

13. The length of time a compound is retained in a chromatographic column is called as
 A) Regression time B) Retention time
 C) Residence time D) Resting time
14. The common carrier gas used in GC is
 A) Oxygen B) Carbon dioxide
 C) Nitrous oxide D) Nitrogen
15. In ion exchange chromatography, separation takes place on the basis of
 A) Mass B) Size
 C) Charge D) Pressure
16. In a GC instrument, TCD stands for
 A) Thermal Conductivity Detector B) Temperature Control Device
 C) The Collected Data D) Top Cooling Device
17. Electrophoresis of RNA is usually carried out on
 A) Acrylamide gels B) Nitrocellulose membranes
 C) Agarose gels D) Paper
18. Sodium dodecyl sulfate (SDS) is a
 A) Acid B) Base
 C) Gel D) Detergent
19. A commonly used gel for protein detection on gels is
 A) Congo Red B) Comassie Brilliant Blue
 C) Malachite Green D) Tryphan Blue
20. In electrophoresis, urea is used as
 A) Buffer B) Gel
 C) Solubilizer D) Dye

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write a note on titration curve.	04	L1	1	1
	b) Calculate the molarity of pure water.	04	L2	1	2
	c) Describe the instrumentation of UV-Vis spectrophotometer.	08	L2	2	1
2.	a) Define ionic strength.	02	L1	1	1
	b) What is calibration? What is its importance? How is single point calibration different from multiple point calibration?	06	L2	1	2
	c) In detail, explain the principle of luminescence spectroscopy.	08	L2	2	2
3.	a) What is the biochemical importance of weak electrolytes?	04	L2	1	1
	b) Describe the instrumentation of Atomic Absorption Spectroscopy.	12	L2	2	2
Unit – II					
4.	a) Based on the principle, explain the importance of IR spectroscopy in analytical measurements.	08	L2	3	2
	b) Discuss about any two types of analyzers used in mass spectroscopy.	08	L2	4	2
5.	a) What is spin-spin coupling effect in NMR? Explain with a suitable example.	08	L2	3	2
	b) Give details on the technique of ion exchange chromatography.	08	L2	4	2
6.	a) Elaborate on MALDI.	06	L2	3	2
	b) Outline the instrumentation of a gas chromatograph.	10	L2	4	2
Unit – III					
7.	a) What is the principle behind electrophoretic separation? How is it carried out?	08	L2	5	2
	b) Discuss about any two types of support media used for electrophoretic separation.	08	L2	5	2
8.	a) Describe the technique of isoelectric focusing in detail.	08	L2	5	2
	b) How is the detection of sample components separated on gels carried out? Explain.	08	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
